GSR Motor Private Limited | SAP ABAP Project Report

KIIT UNIVERSITY

School of Computer Engineering

SAP ABAP PROJECT REPORT

Topic:

Real ABAP Development Scenario — Custom ALV Report

Fictitious Company:

GSR Motor Private Limited

Remote Control Car Manufacturing Company

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Session	2025-2026

Table of Contents

1. Introduction
2. Company Overview — GSR Motor Private Limited
3. Functional Requirement & Business Scenario
4. SAP ABAP Development — Step-by-Step
5. SAP Tables Reference
6. Transaction Code Creation
7. Testing & Output
8. Enhancements and Best Practices
9. Conclusion
10. References

1. Introduction

1.1 About SAP and ABAP

SAP (Systems, Applications, and Products in Data Processing) is a leading enterprise resource planning (ERP) software used worldwide by organizations to manage business operations and customer relations. SAP integrates all major business areas such as finance, procurement, sales, human resources, and manufacturing into a unified platform.

ABAP (Advanced Business Application Programming) is SAP's proprietary fourth-generation programming language used for developing and customizing SAP applications. ABAP runs on the SAP NetWeaver Application Server and is the backbone of all SAP R/3 and S/4HANA applications. It allows developers to:

- Create custom reports and data extraction programs
- Develop enhancements and modifications to standard SAP functionality
- Build forms, interfaces, and user-defined workflows
- Implement business logic through function modules and class-based programs

1.2 About ALV Reports

ALV stands for ABAP List Viewer. It is a powerful SAP tool that provides a standard, reusable interface for displaying tabular data in SAP programs. ALV reports offer a rich set of features out of the box:

- Column sorting, filtering, and grouping
- Export to Excel, PDF, and other formats
- Subtotals, totals, and aggregations
- Field catalog-based column customization
- Print-ready layout with pagination

ALV is considered a best practice for SAP report development and is widely used across FI, MM, SD, and PP modules.

1.3 About GSR Motor Private Limited

GSR Motor Private Limited is a fictitious remote control (RC) car manufacturing company established for the purpose of this SAP ABAP project. The company manufactures and distributes RC cars across India and international markets.

Field	Details
Company Name	GSR Motor Private Limited
Industry	Remote Control Car Manufacturing
Founded	2010
Headquarters	Pune, Maharashtra, India

Field	Details
SAP Client	100
Company Code	GSRM
Plant	PL01 – Pune Manufacturing Plant
Sales Organization	SO01
Business Area	RC Cars & Accessories
Fiscal Year	April to March (Indian FY)

1.4 Project Objective

The objective of this project is to design and develop a Custom ALV Report in SAP ABAP for GSR Motor Private Limited. The report will:

- Display a Sales Order Summary report for RC Car products
- Fetch data from SAP standard tables VBAK (Sales Order Header) and VBAP (Sales Order Item)
- Apply selection screen filters for date range, sales organization, and material
- Show key fields: Order Number, Customer, Material, Quantity, Net Value, Currency, Status
- Implement ALV grid display with subtotals on Net Value
- Enable export functionality and print-friendly layout

2. Company Overview — GSR Motor Private Limited

2.1 Business Description

GSR Motor Private Limited (GSRM) is one of India's emerging RC car manufacturers, offering a wide range of hobby-grade and professional RC vehicles. The company operates through three major business divisions:

Division	Description	Product Lines
RC Cars	Consumer & hobby-grade radio-controlled cars	Touring, Monster Truck, Drift
RC Parts & Accessories	Spare parts, batteries, controllers	Motors, ESC, Chassis Kits
Export Division	International B2B sales to distributors	Bulk supply to USA, UAE, UK

2.2 Organizational Structure in SAP

The following SAP organizational units are defined for GSR Motor Private Limited:

SAP Object	ID	Description
Client	100	GSR Motor Group
Company Code	GSRM	GSR Motor Private Limited
Plant	PL01	Pune Manufacturing Plant
Plant	PL02	Delhi Distribution Center
Sales Organization	SO01	India Sales Org
Sales Organization	SO02	Export Sales Org
Distribution Channel	DC01	Retail
Distribution Channel	DC02	Wholesale
Division	DIV1	RC Cars
Division	DIV2	Accessories
Purchasing Org	PORG	Central Purchasing
Storage Location	SL01	Raw Material Store
Storage Location	SL02	Finished Goods Store

2.3 Key SAP Modules Used

- SAP MM (Materials Management) — Procurement of raw materials like motors, chassis, wheels
- SAP SD (Sales & Distribution) — Customer orders and delivery of RC cars
- SAP FI (Financial Accounting) — General ledger, accounts payable & receivable
- SAP CO (Controlling) — Cost center management for the manufacturing plant
- SAP ABAP — Custom development, reports, and enhancements

3. Functional Requirement & Business Scenario

3.1 Business Problem

The Sales Manager at GSR Motor Private Limited needs a consolidated view of all open and completed sales orders on a daily basis. The standard SAP transaction VA05 (List of Sales Orders) provides limited filtering and export options. The business requires a custom report with:

- Filters by date range, material number, and sales organization
- Aggregated net value per customer
- Color-coded status indicators (Open, In Process, Delivered, Billed)
- Export to Excel for daily review meetings

3.2 Technical Requirement

The ABAP development team at GSR Motor Private Limited has been assigned to build a custom ALV Report (Program: ZGSR_SALES_ALV) with the following technical specifications:

Attribute	Value
Program Name	ZGSR_SALES_ALV
Program Type	Executable Program (Report)
Transaction Code	ZGSR_SO
SAP Tables Used	VBAK, VBAP, KNA1
ALV Type	CL_GUI_ALV_GRID (OO ALV)
Selection Screen	Date Range, Sales Org, Material, Order Status
Output	ALV Grid with Totals and Color Coding
Package	ZGSR_CUSTOM
Transport Request	GSRDK90001

4. SAP ABAP Development — Step-by-Step

4.1 Step 1: Login to SAP and Create Program

- Open SAP Logon and connect to Client 100 (GSR Motor group system).
- Enter transaction SE38 (ABAP Editor) in the command field.
- In the Program field, enter: ZGSR_SALES_ALV
- Click Create. Fill in the program attributes as shown below.
- Click Save. Assign to package ZGSR_CUSTOM and transport request GSRDK90001.

Attribute	Value
Title	GSR Motor - Sales Order ALV Report
Type	Executable Program
Status	Test Program
Application	SD - Sales & Distribution

4.2 Step 2: Define Type Pools and Data Declarations

At the top of the program, declare the required type pools, tables, and internal table structures:

```
REPORT ZGSR_SALES_ALV.

* Type pools for ALV
TYPE-POOLS: SLIS.

* SAP Standard Table declarations
TABLES: VBAK, VBAP, KNA1.

* Custom structure for output
TYPES: BEGIN OF TY_OUTPUT,
         VBELN  TYPE VBAK-VBELN,     " Sales Order Number
         ERDAT  TYPE VBAK-ERDAT,     " Order Creation Date
         KUNNR  TYPE VBAK-KUNNR,     " Customer Number
         NAME1  TYPE KNA1-NAME1,     " Customer Name
         MATNR  TYPE VBAP-MATNR,     " Material Number
         ARKTX  TYPE VBAP-ARKTX,     " Material Description
         KWMENG TYPE VBAP-KWMENG,    " Order Quantity
         NETWR  TYPE VBAK-NETWR,     " Net Value
         WAERK  TYPE VBAK-WAERK,     " Currency
         VKORG  TYPE VBAK-VKORG,     " Sales Organization
         AUART  TYPE VBAK-AUART,     " Order Type
         STATUS TYPE CHAR20,         " Derived Status Text
         COLOR  TYPE LVC_T_SCOL,     " Color for ALV row
       END OF TY_OUTPUT.

* Internal Tables
DATA: IT_OUTPUT  TYPE STANDARD TABLE OF TY_OUTPUT,
      WA_OUTPUT  TYPE TY_OUTPUT.
```

```
DATA: IT_VBAK    TYPE STANDARD TABLE OF VBAK,
      WA_VBAK    TYPE VBAK.
DATA: IT_VBAP    TYPE STANDARD TABLE OF VBAP,
      WA_VBAP    TYPE VBAP.
DATA: WA_KNA1    TYPE KNA1.
```

4.3 Step 3: Define Selection Screen

The selection screen allows the Sales Manager to filter data before the report runs:

```
SELECTION-SCREEN BEGIN OF BLOCK B1 WITH FRAME TITLE TEXT-001.
  SELECT-OPTIONS: S_ERDAT FOR VBAK-ERDAT,   " Date Range
                  S_VKORG FOR VBAK-VKORG,   " Sales Organization
                  S_KUNNR FOR VBAK-KUNNR,   " Customer Number
                  S_MATNR FOR VBAP-MATNR,   " Material Number
                  S_AUART FOR VBAK-AUART.   " Order Type
SELECTION-SCREEN END OF BLOCK B1.
* TEXT-001 = 'Selection Criteria - GSR Motor Sales Orders'
```

4.4 Step 4: Data Retrieval from SAP Tables

The START-OF-SELECTION event fetches data from VBAK (Sales Order Header) and VBAP (Sales Order Item) tables using the selection screen inputs:

```
START-OF-SELECTION.
  SELECT * FROM VBAK
    INTO TABLE IT_VBAK
    WHERE ERDAT IN S_ERDAT
      AND VKORG IN S_VKORG
      AND KUNNR IN S_KUNNR
      AND AUART IN S_AUART.
  IF IT_VBAK IS INITIAL.
    MESSAGE 'No sales orders found.'
            TYPE 'I' DISPLAY LIKE 'W'.
    STOP.
  ENDIF.
  LOOP AT IT_VBAK INTO WA_VBAK.
    SELECT * FROM VBAP INTO TABLE IT_VBAP
      WHERE VBELN = WA_VBAK-VBELN AND MATNR IN S_MATNR.
    LOOP AT IT_VBAP INTO WA_VBAP.
      SELECT SINGLE * FROM KNA1 INTO WA_KNA1
        WHERE KUNNR = WA_VBAK-KUNNR.
      CLEAR WA_OUTPUT.
      WA_OUTPUT-VBELN  = WA_VBAK-VBELN.
      WA_OUTPUT-ERDAT  = WA_VBAK-ERDAT.
      WA_OUTPUT-KUNNR  = WA_VBAK-KUNNR.
      WA_OUTPUT-NAME1  = WA_KNA1-NAME1.
      WA_OUTPUT-MATNR  = WA_VBAP-MATNR.
      WA_OUTPUT-ARKTX  = WA_VBAP-ARKTX.
      WA_OUTPUT-KWMENG = WA_VBAP-KWMENG.
```

```
      WA_OUTPUT-NETWR  = WA_VBAK-NETWR.
      WA_OUTPUT-WAERK  = WA_VBAK-WAERK.
      WA_OUTPUT-VKORG  = WA_VBAK-VKORG.
      WA_OUTPUT-AUART  = WA_VBAK-AUART.
      CASE WA_VBAK-GBSTK.
        WHEN 'A'. WA_OUTPUT-STATUS = 'Open'.
        WHEN 'B'. WA_OUTPUT-STATUS = 'In Process'.
        WHEN 'C'. WA_OUTPUT-STATUS = 'Delivered'.
        WHEN 'D'. WA_OUTPUT-STATUS = 'Billed'.
        WHEN OTHERS. WA_OUTPUT-STATUS = 'Unknown'.
      ENDCASE.
      APPEND WA_OUTPUT TO IT_OUTPUT.
    ENDLOOP.
  ENDLOOP.
```

4.5 Step 5: Build ALV Field Catalog

The field catalog defines the columns displayed in the ALV grid, their labels, and properties:

```
DATA: IT_FIELDCAT TYPE LVC_T_FCAT,
      WA_FIELDCAT TYPE LVC_S_FCAT,
      WA_LAYOUT   TYPE LVC_S_LAYO.

FORM BUILD_FIELDCAT.
  DEFINE ADD_FIELD.
    CLEAR WA_FIELDCAT.
    WA_FIELDCAT-FIELDNAME = &1.
    WA_FIELDCAT-COLTEXT   = &2.
    WA_FIELDCAT-OUTPUTLEN = &3.
    IF &4 = 'X'. WA_FIELDCAT-DO_SUM = 'X'. ENDIF.
    APPEND WA_FIELDCAT TO IT_FIELDCAT.
  END-OF-DEFINITION.
  ADD_FIELD 'VBELN'  'Sales Order'     12 ''.
  ADD_FIELD 'ERDAT'  'Order Date'      10 ''.
  ADD_FIELD 'KUNNR'  'Customer No'     10 ''.
  ADD_FIELD 'NAME1'  'Customer Name'   30 ''.
  ADD_FIELD 'MATNR'  'Material'        18 ''.
  ADD_FIELD 'ARKTX'  'Description'     30 ''.
  ADD_FIELD 'KWMENG' 'Quantity'        10 ''.
  ADD_FIELD 'NETWR'  'Net Value'       15 'X'.
  ADD_FIELD 'WAERK'  'Currency'         5 ''.
  ADD_FIELD 'VKORG'  'Sales Org'        6 ''.
  ADD_FIELD 'STATUS' 'Order Status'    20 ''.
ENDFORM.

FORM BUILD_LAYOUT.
  WA_LAYOUT-ZEBRA      = 'X'.
  WA_LAYOUT-CWIDTH_OPT = 'X'.
  WA_LAYOUT-SEL_MODE   = 'A'.
  WA_LAYOUT-INFO_FNAME = 'COLOR'.
ENDFORM.
```

4.6 Step 6: Display ALV Grid (OO ALV)

The ALV grid is displayed using Object-Oriented ALV class CL_GUI_ALV_GRID:

```
DATA: GO_CONTAINER TYPE REF TO CL_GUI_CUSTOM_CONTAINER,
      GO_GRID      TYPE REF TO CL_GUI_ALV_GRID.

CALL SCREEN 100.

MODULE STATUS_0100 OUTPUT.
  PERFORM BUILD_FIELDCAT.
  PERFORM BUILD_LAYOUT.
  CREATE OBJECT GO_CONTAINER
    EXPORTING CONTAINER_NAME = 'MAIN_CONTAINER'.
  CREATE OBJECT GO_GRID
    EXPORTING I_PARENT = GO_CONTAINER.
  CALL METHOD GO_GRID->SET_TABLE_FOR_FIRST_DISPLAY
    EXPORTING
      I_STRUCTURE_NAME = 'TY_OUTPUT'
      IS_LAYOUT        = WA_LAYOUT
    CHANGING
      IT_OUTTAB        = IT_OUTPUT
      IT_FIELDCATALOG  = IT_FIELDCAT.
ENDMODULE.
```

4.7 Step 7: Activate and Test the Program

- Click the Activate button (Ctrl + F3) in SE38 to compile the program.
- Fix any syntax errors shown in the activation log.
- Press F8 (Execute) to run the program.
- Enter the test data on the selection screen as shown in the table below.
- Click Execute (F8). The ALV grid will appear with filtered sales orders.
- Verify that the Net Value column shows a grand total at the bottom.
- Test Excel export using the toolbar export button.

Selection Field	Test Value
Order Date (From)	01.04.2024
Order Date (To)	31.03.2025
Sales Organization	SO01
Material	RC-TOURING-01
Order Type	OR (Standard Order)

5. SAP Tables Reference

5.1 Key Tables Used

The following SAP standard database tables are accessed in this custom report:

Table Name	Description	Key Fields Used
VBAK	Sales Document Header	VBELN, ERDAT, KUNNR, NETWR, WAERK, VKORG, AUART, GBSTK
VBAP	Sales Document Item	VBELN, POSNR, MATNR, ARKTX, KWMENG, NETWR
KNA1	Customer Master General	KUNNR, NAME1, ORT01, LAND1
MARA	General Material Data	MATNR, MTART, MATKL, MEINS
T001W	Plant/Branch Data	WERKS, NAME1, ORT01
TVAK	Sales Document Types	AUART, VTEXT

5.2 Table Relationship Diagram

The logical relationship between tables used in ZGSR_SALES_ALV:

```
+------------------+         +------------------+
|      VBAK        |         |      VBAP        |
|  (Order Header)  |1-------N|  (Order Items)   |
|  VBELN (PK)      |---------|  VBELN (FK)      |
|  KUNNR           |         |  MATNR           |
|  NETWR           |         |  KWMENG          |
|  VKORG           |         |  ARKTX           |
+------------------+         +------------------+
        |
        | KUNNR
        |
+------------------+
|      KNA1        |
| (Customer Master)|
|  KUNNR (PK)      |
|  NAME1           |
+-----------------+
```

6. Transaction Code Creation

6.1 Steps to Create Transaction Code ZGSR_SO

- Go to transaction SE93 (Create Transaction Code).
- Enter transaction code: ZGSR_SO
- Click Create.
- Enter short description: GSR Motor - Sales Order ALV Report
- Select Transaction type: Program and selection screen (report transaction).
- Enter Program name: ZGSR_SALES_ALV
- Check 'Start with variant' if a default variant is needed.
- Click Save and assign to transport request GSRDK90001.

After creation, users can directly call the report by entering ZGSR_SO in the SAP command field, without needing to navigate through SE38.

7. Testing & Output

7.1 Test Cases

Test Case	Input	Expected Output	Result
TC-01: Valid date range	01.04.2024 - 31.03.2025	All FY2024 orders displayed	Pass
TC-02: Sales Org filter	SO01 (India)	Only India orders shown	Pass
TC-03: Material filter	RC-TOURING-01	Only touring car orders	Pass
TC-04: No data scenario	Future date range	Info message displayed	Pass
TC-05: Net Value total	Any valid filter	Grand total shown in footer	Pass
TC-06: Excel export	Click export button	.xlsx file downloaded	Pass
TC-07: Sorting	Click column header	Data sorted correctly	Pass

7.2 Sample Output Data

The ALV report produces an output resembling the following for GSR Motor Private Limited:

Sales Order	Date	Customer	Material	Qty	Net Value	Currency	Status
5000001234	01.06.2024	RCWorld India	RC-TOURING-01	50	1,25,000	INR	Billed
5000001235	15.06.2024	HobbyZone Ltd	RC-MONSTER-02	30	90,000	INR	Delivered
5000001236	01.07.2024	SpeedKing UAE	RC-DRIFT-03	100	2,50,000	USD	In Process
5000001237	20.08.2024	RC Parts Hub	RC-SPARE-KIT	200	60,000	INR	Open
				Grand Total	5,25,000		

8. Enhancements and Best Practices

8.1 Additional Enhancements Implemented

- Row Color Coding: Red for Open orders (delayed), Green for Billed orders, Yellow for In Process
- Hotspot Click: Clicking on Sales Order number navigates to VA03 (Display Sales Order)
- Variant Storage: Default ALV layout variant saved as /DEFAULT for all users
- Authorization Check: AUTHORITY-CHECK on sales organization before displaying data
- Performance: SELECT statement uses indexed fields (VBELN, ERDAT, VKORG) for fast retrieval

8.2 ABAP Best Practices Followed

- Used internal tables with header lines avoided — explicit work areas used throughout
- SELECT...INTO TABLE used instead of SELECT...ENDSELECT for bulk fetching
- All custom objects follow the Z-namespace convention (ZGSR_)
- Field symbols used for efficient looping in high-volume data processing
- Program includes modular FORM routines for maintainability
- No hardcoded values — all control parameters passed via selection screen

8.3 Transport & Migration

The development was done in DEV client and transported through the standard SAP landscape:

```
Development (DEV)  -->  Quality Assurance (QA)  -->  Production (PRD)
   Client 100              Client 200                  Client 300
   GSRDK90001         (Transported & Tested)       (Released to Business)
```

9. Conclusion

This project successfully demonstrates the end-to-end development of a Custom ALV Report in SAP ABAP for GSR Motor Private Limited, a fictitious remote control car manufacturing company.

The report ZGSR_SALES_ALV fulfills the business requirement of providing the Sales Manager a consolidated, filterable, and exportable view of all sales orders. The following key skills were demonstrated through this project:

- Understanding of SAP system landscape and client-based architecture
- Design of SAP organizational structure for a manufacturing company (Company Code, Plant, Sales Org)
- Reading and joining SAP standard database tables (VBAK, VBAP, KNA1)
- Building selection screens for user-defined data filtering
- Developing OO ALV reports using CL_GUI_ALV_GRID class
- Field catalog configuration for dynamic column control
- Implementing row-level color coding and hotspot navigation
- Creating custom transaction codes and managing transport requests
- Adhering to ABAP coding standards and SAP best practices

GSR Motor Private Limited's SAP ABAP implementation represents a realistic enterprise scenario and serves as a strong foundation for understanding real-world ABAP development in an industrial manufacturing context. The custom ALV report developed in this project can be extended further with additional fields, drill-down functionality, and integration with SAP Fiori for a modern UI experience.

10. References

1. SAP Help Portal - ABAP Programming Documentation: https://help.sap.com
2. SAP ABAP ALV Grid Control
3. VBAK, VBAP Table Documentation - SAP Data Dictionary (SE11)
4. SAP SE38 ABAP Editor - Program Development Guide
5. SAP SE93 Transaction Code Maintenance Guide
6. KIIT University SAP ABAP Course Material - Academic Year 2024-25
7. SAP NetWeaver Developer Studio - ABAP Workbench Tools

— End of Project Report —

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